

Project 5 Notations & Summary

Project 5 Notations

Notation	Meaning	Definition	Description
H	Hidden states		The set of all possible game states (not user observable.)
a	An action		
a_t	Action chosen at time t		
b	An object		
b_t	Object chosen at time t		
C	Commands	$\{(a,b)\}$	The set of all commands (action-object) pairs.
c	A command	$c = (a, b)$	
P	Transition probabilities	$P : H \times C \times H \rightarrow [0, 1]$	$P(h' h, a, b)$ is the probability of reaching state h' by taking command (a, b) in state h .
R	Reward function	$R : H \times C \rightarrow \mathbb{R}$	$R(h, a, b)$ is the reward of taking command (a, b) in state h .
γ	Discount factor		
S	Observable states	$\{s\}$	The set of all possible text descriptions.
Ψ	Observation state generator	$\Psi : H \rightarrow S$	A stochastic function used to produce text descriptions s observable by the player.
r	Room index		
q	Quest index		Example of a quest: <i>You are hungry now.</i>
Γ_0	Initial state distribution		uniform over H .
h	A hidden state	$h = (r, q) \in H$	
$h(s)$	The uniquely associated hidden state of an observable state s	$h : S \rightarrow H$	Getting the hidden state from an observable state, $h(s)$, is deterministic, whereas getting an observable state from a hidden state, $\Psi(h)$, is stochastic.
h_t	Hidden state at time t	$h_t = (h_{r,t}, h_{q,t})$	$h_t \sim P(\cdot h_{t-1}, a_{t-1}, b_{t-1})$ and $h_0 = (h_{r,0}, h_{q,0}) \sim \Gamma_0$
$h_{r,t}$	The room index/component of a hidden state at time t		
$h_{q,t}$	The quest index/component of a hidden state at time t	$h_{q,t} = h_{q,0}$	The quest remains unchanged.
s_r	Room description		Randomly provided.
s_q	Quest description		
s	An observable state	$s = (s_r, s_q) \in S, h(s) \in H$	Associated with a unique hidden state.
s_t	An observable state at time t	$s_t \sim \Psi(h_t)$	
$s_{q,t}$	The quest component of an observable state at time t	$s_{q,t} = s_{q,0}$	The quest remains unchanged.
K	Time at which player finishes the quest.		
r_t	Reward at time t	$r_t = R(h_t, a_t, b_t)$	$r_t = 0$ for $t > K$.

Notation	Meaning	Definition	Description
$V^*(s)$	Maximum expected total discounted reward from a given observable starting state s over all possible policies.	(see summary)	
V^*	The weighted average of $V^*(s)$ over all possible starting states.	(see summary)	Weighted average of all the maximum expected total discounted reward across all possible starting states.
$Q^*(s, a, b)$	Optimal Q-function	(see summary)	Maximum expected total discounted reward from a given observable starting state s , taking the specific action (a, b) at state s , over all possible policies.
π	Policy	$\pi : S \rightarrow C$	Determine the command given an observable state.
$\pi^*(s)$	Optimal policy/command at state s	(see summary)	The specific command (a, b) at state s that would yield the maximum expected total discounted reward starting from state s , i.e., $Q^*(s, a, b)$

Summary

- Full record of interaction:

$$(h_0, s_0, a_0, b_0, r_0, \dots, h_t, s_t, a_t, b_t, r_t, h_{t+1}, \dots)$$

- Observable record of interaction:

$$(s_0, a_0, b_0, r_0, \dots, s_t, a_t, b_t, r_t, \dots)$$

- Total discounted reward:

$$\sum_{t=0}^{\infty} \gamma^t r_t$$

- Expected cumulative discounted reward:

$$\mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(h_t, a_t, b_t) \mid (a_t, b_t) \sim \pi \right]$$

- Maximum expected total discounted reward starting from a given observable state s over all possible policies:

$$V^*(s) = \max_{\pi} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(h_t, a_t, b_t) \mid h_0 = h(s), s_0 = s, (a_t, b_t) \sim \pi \right]$$

- The weighted average of $V^*(s)$ over all possible starting states:

$$V^* = \mathbb{E}_{h_0 \sim \Gamma_0, s \sim \Psi(h)} [V^*(s)]$$

- Maximum expected total discounted reward from a given observable starting state s , taking the specific action (a, b) at state s at time $t = 0$, over all possible policies.

$$Q^*(s, a, b) = \max_{\pi} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(h_t, a_t, b_t) \mid h_0 = h(s), s_0 = s, a_0 = a, b_0 = b, (a_t, b_t) \sim \pi \text{ for } t \geq 1 \right]$$

- Optimal policy/command at state s

$$\pi^*(s) = \arg \max_{(a,b) \in C} Q^*(s, a, b)$$

Clarifying notes

- s_t is sampled from h_t via $\Psi(h_t)$ at every time step. So even if $h_t = h_{t+1}$, s_t may not equal s_{t+1} .
- The reason of $(a_t, b_t) \sim \pi$ is because even though the policy function π is deterministic given s_t , the observable state s_t itself is randomly generated from h_t via the stochastic function Ψ , and h_t itself is randomly generated from the transition probability function P if $t > 0$ or Γ_0 if $t = 0$.
- Γ_0 and P are fixed, but they are not directly observable by the player.
- V^* is the weighted average of $V^*(s)$ over all possible starting states — even though Γ_0 is uniform, Ψ is not necessarily uniform.